Quantitative Aptitude Formula Handbook

Prepo by Avneesh — Core Formulas & Shortcuts | Category: FORMULA SHEET

1. Percentages, Profit & Loss

- Fraction to %: 1/2 = 50%, 1/3 = 33.33%, 1/4 = 25%, 1/5 = 20%, 1/6 = 16.66%, 1/7 = 14.28%, 1/8 = 12.5%
- Profit % = (Profit / CP) * 100 | Loss % = (Loss / CP) * 100
- Selling Price (SP) = CP * (1 + Profit% / 100) or CP * (1 - Loss% / 100)
- Successive Discounts: Single discount equivalent to d1% and d2% = [d1 + d2 - (d1 * d2) / 100]%
- Faulty Weights: True Weight / False Weight = (100 + Gain%) / 100

2. Time & Work, Pipes & Cisterns

- Work = Efficiency * Time | If A does work in X days and B in Y days: Together = (X * Y) / (X + Y)
- Men, Days, Hours formula: (M1 * D1 * H1) / W1 = (M2 * D2 * H2) / W2
- Alternating Work: Calculate work completed in cycles of 2 or 3 days first to avoid overshoot.
- Inlet / Outlet Pipes: Net Rate = Rate of Inlets - Rate of Outlets

3. Time, Speed & Distance, Trains & Boats

- Conversion: 1 km/h = 5/18 m/s | 1 m/s = 18/5 km/h
- Average Speed (Equal distances at speed A and B) = 2AB / (A + B)
- Relative Speed (Same direction) = S1 - S2 | (Opposite direction) = S1 + S2
- Crossing a platform of length P by train of length L: Total Distance = L + P
- Boats: Downstream = Speed + Current | Upstream = Speed - Current

4. Simple Interest & Compound Interest

- Simple Interest (SI) = (P * R * T) / 100
- Compound Amount = P * (1 + R / 100)^T
- Difference between CI and SI for 2 years = P * (R / 100)^2
- Difference between CI and SI for 3 years = P * (R / 100)^2 * (3 + R / 100)